

Auteur: Joe Ayoub
Studierichting: Informatica
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$$\begin{aligned} I &= \int (x^2 - x - 4) \cdot e^x dx \\ \begin{cases} u = x^2 - x - 4 \\ dv = e^x dx \end{cases} &\Rightarrow \begin{cases} du = (2x - 1)dx \\ v = e^x \end{cases} \\ I &= (x^2 - x - 4)e^x - \int e^x(2x - 1)dx \\ \begin{cases} u = 2x - 1 \\ dv = e^x dx \end{cases} &\Rightarrow \begin{cases} du = 2dx \\ v = e^x \end{cases} \\ I &= (x^2 - x - 4)e^x - \left((2x - 1)e^x - 2 \int e^x dx \right) \\ &= (x^2 - x - 2)e^x - ((2x - 1)e^x - 2e^x) + C \\ &= (x^2 - 3x - 1)e^x + C \end{aligned}$$

References